On Node Ranking of Graphs under Strong Orientation

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Abstract

For a graph $G = (V, E)$, a mapping $C : V \to \{1, 2, \cdots k\}$ is a node ranking labeling if every path between any two vertices u and v, with $C(u) = C(v)$, there is a node w on the path with $C(w) > C(u) = C(v)$. The minimum possible k such that the node ranking labeling of G exists is the node ranking number of G. The node ranking problem is the problem of finding the node ranking number of graphs. The directed node ranking problem may be defined accordingly. The upper/lower node ranking number of a graph G is the maximum/minimum possible directed node ranking number among all strong orientations of G. This paper established the upper and lower node ranking number of wheel graphs.

1 Introduction

A graph $G = (V, E)$ is *k-rankable* if there is a mapping $C : V \to \{1, 2, \cdots k\}$ such that for every *u-v* path of *G* with $C(u) = C(v)$, there is a vertex *w* lies on the path such that $C(w) > C(u) = C(v)$. The mapping *C* is called a *node ranking labeling of G* and the value $C(v)$ is called the *rank* or *color* of the vertex *v*. The *node ranking number* $\chi_r(G)$ of a graph *G* is the smallest integer *k* such that *G* is *k*-rankable. A node ranking labeling is called an *optimal node ranking labeling of G* if its maximum rank is $\chi_r(G)$. The *node ranking problem* (also called *ordered coloring problem* [7]) for a graph *G* is the problem of finding the node ranking number $\chi_r(G)$ of *G*. For an arbitrary graph *G*, the decision version of the node ranking problem is NP-complete [1]. In fact, this problem remains NP-Complete even with restriction of the graph to be co-bipartite graphs [12]. Since the node ranking number of a graph *G* is equivalent to the minimum height of the separation tree on *G* [3, 4], finding the node ranking number is important when applying on communication network design [3, 6, 11, 14], computing Cholesky factorizations of matrices in parallel [1, 5, 10], and VLSI layout problem [9, 15] etc.

Node ranking problem has been studied since 1980s. It is known that $\chi_r(P_n) = \lfloor \log n \rfloor + 1$ for $n \geq 1$ [7] and $\chi_r(C_n) = \lfloor \log(n-1) \rfloor + 2$ for $n \geq 3$ [2]. An upper bound of the node ranking number of an arbitrary tree with *n* nodes was given by [6], they proposed an $O(n \log n)$ time optimal node ranking labeling algorithm of trees, which was further improved to $O(n)$ by [13].

We may define the node ranking problem on digraph accordingly. A digraph $D = (V, E)$ is *k-rankable* if there is a mapping $C' : V \to \{1, 2, \cdots k\}$ such that for every *u-v* directed path of *D* with $C'(u) = C'(v)$. There is a vertex *w* on the path with $C'(w) > C'(u) = C'(v)$. The mapping C' is called a (*directed*) *node ranking labeling of D* and the value $C'(v)$ is called the *rank* or *color* of the vertex *v*. The (*directed*) *ranking number* $\chi'_r(D)$ of a graph *D* is the smallest integer *k* such that *D* is *k*-rankable. Since not every pair of vertices have a directed path in digraph *D*, we only consider strong digraphs. Define the upper node ranking number among all strong orientation of a graph *G* as $\text{RANK}(G) = \max\{\chi'_r(D) : D \text{ is a strong orientation of } G\}$. Similarly, the lower node ranking number among all strong orientation of *G* is defined as $\text{rank}(G) = \min\{\chi'_r(D) : D \text{ is a strong orientation of } G\}$. This paper discuss the upper and lower node ranking number of wheel graphs. Let $W_{1,n}$ denote a wheel graph with a vertex in the center and *n* vertices form a cycle of the wheel. Figure 1 shows an example of an optimal node ranking labeling of $W_{1,8}$ and shows different directed node ranking labelings of a strong orientation of $W_{1,8}$.

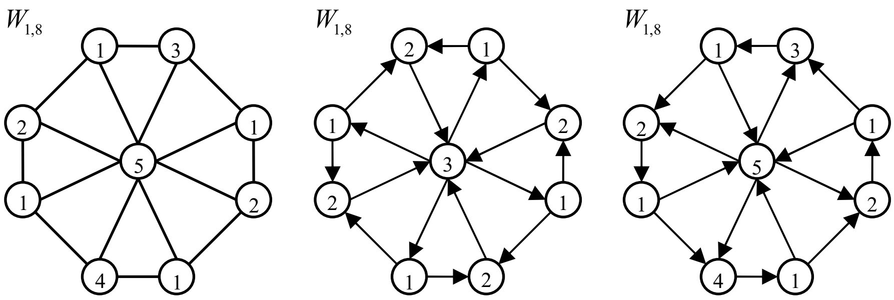

Figure 1: Node ranking labelings on undirected and strong orientations of $W_{1,8}$.

2 Main results

Lemma 1: [2] *The node ranking number of a cycle* C_n *is* $\chi_r(C_n) = \lceil \lg n \rceil + 1$.

Since the only strong orientation of a cycle is either a clockwise or counterclockwise directed cycle, by lemma 1, theorem 2 comes trivial.

Lemma 2: *Let* $G = C_n$ *be a cycle on n vertices. Then the upper and lower node ranking number* $RANK(G) = rank(G) = \lceil \lg n \rceil + 1$.

Given *H* be a subgraph of *G*, since every *u*-*v* path in *H* must be a *u*-*v* path in *G*, but not the reverse, we have following proposition:

Proposition 1: *Let H be a subgraph of G. Then* $\chi_r(H) \le \chi_r(G)$.

Similar to the undirected graph, digraphs have the same proposition which is stated as proposition 2.

Proposition 2: *Let H be a subgraph of D. Then* $\chi'_r(H) \le \chi'_r(D)$.

Lemma 3: *The node ranking number of a wheel* $W_{1,n}$ *is* $\chi_r(W_{1,n}) = \lceil \lg n \rceil + 2$.

Proof：Let v_c denote the center vertex of $W_{1,n}$ and $V_n = \{v_1, v_2, ..., v_n\}$ denote the vertices on the cycle of the wheel. To see that $\chi_r(W_{1,n}) \ge \lceil \lg n \rceil + 2$, suppose to the contrary, $\chi_r(W_{1,n}) \le \lceil \lg n \rceil + 1$. Let $C : V(W_{1,n}) \to \{1, 2, \cdots, \lceil \lg n \rceil + 1\}$ be an optimal node ranking labeling of $W_{1,n}$. Since v_c is adjacent to v_i for all $v_i \in V_n$, we have $C(v_c) \ne C(v_i), \forall v_i \in V_n$. Since every pair of vertices v_i, v_j with $C(v_i) = C(v_j)$, v_i-v_c-v_j is a path in $W_{1,n}$, without loss of generality, we may have $C(v_c) = \lceil \lg n \rceil + 1$ which implies $C(v_i) \le \lceil \lg n \rceil$ for all $v_i \in V_n$. Consider the subgraph $C_n = W_{1,n} - v_c$, let $C' : V(C_n) \to \{1, 2, \cdots, \lceil \lg n \rceil\}$ be a labeling defined as $C'(v_i) = C(v_i)$ for all $v_i \in V_n$. Then C' is a node ranking labeling of C_n with maximum rank $\lceil \lg n \rceil$, which implies $\chi_r(C_n) \le \lceil \lg n \rceil$ contradicts to Lemma 1. Hence $\chi_r(W_{1,n}) \ge \lceil \lg n \rceil + 2$. Now consider a labeling $f : V(W_{1,n}) \to \{1, 2, \cdots \lceil \lg n \rceil + 2\}$ which labels $V_n = \{v_1, v_2, ..., v_n\}$ by an optimal node ranking labeling of C_n and labels the vertex v_c as $\lceil \lg n \rceil + 2$. Then *f* is a node ranking labeling of $W_{1,n}$ with maximum rank $\lceil \lg n \rceil + 2$ which implies $\chi_r(W_{1,n}) \le \lceil \lg n \rceil + 2$. Hence we have $\chi_r(W_{1,n}) = \lceil \lg n \rceil + 2$. ■

Since each directed path *P* in a strong orientation *D* of graph *G* is a path in *G*, if *f* is an optimal node ranking of *D* then *f* must be a node ranking labeling of *G*. Hence we have proposition 3 as follows.

Proposition 3: *Let G be a graph and D be an orientation of G. Then* $\chi'_r(D) \le \chi_r(G)$.

Since the adjacent vertices may not have the same rank, lemma 4 is trivial.

Lemma 4: *The node ranking number of complete graph is* $\chi_r(K_n) = n$. *For any tournament D with n vertices,* $\chi'_r(D) = n$.

Lemma 5: [8] *Le G be a graph* $C : V(G) \to \{1, 2, \cdots k\}$ *be an optimal node ranking labeling of* G. *Then* $\forall i, 1 \le i \le k$, *there is a vertex* $v \in V(G)$ *such that* $C(v) = i$.

Theorem 1: *Let* $G = W_{1,n}$ *be a wheel graph. Then* $RANK(G) = \lceil \lg n \rceil + 2$.

Proof: Similar to Lemma 3, let v_c denote the center vertex of *G* and $V_n = \{v_1, v_2, ..., v_n\}$ denote the suburb vertices which form a cycle on the wheel. By lemma 3 and proposition 3, we have $RANK(G) \le \lceil \lg n \rceil + 2$. Consider an orientation *D* of *G* which makes $V_n = \{v_1, v_2, ..., v_n\}$ to be a directed cycle and let v_c have both out-neighbors and in-neighbors, then *D* is a strong orientation of *G*. Consider the same labeling *f* as in lemma 3 on *D*, it is a node ranking labeling of *D* with maximum rank $\lceil \lg n \rceil + 2$ which implies $\chi_r'(D) \le \lceil \lg n \rceil + 2$. By lemma 2 and lemma 5, since $V_n[D]$ is a directed cycle which has at least $\lceil \lg n \rceil + 1$ ranks and v_c is adjacent with v_i for all $v_i \in V_n$, for any directed node ranking labeling C' of *D*, we must have $C'(v_c) \ne C'(v_i), \forall v_i \in V_n$, which implies C' must have at least $\lceil \lg n \rceil + 2$ ranks. Then we have $RANK(G) \ge \chi_r'(D) \ge \lceil \lg n \rceil + 2$. Hence, $RANK(G) = \lceil \lg n \rceil + 2$. ■

Theorem 2: *Let* $G = W_{1,n}$ *be a wheel graph. Then* $rank(G) = \begin{cases} 3 & \text{if } n \text{ is even,} \\ 4 & \text{if } n \text{ is odd.} \end{cases}$

Proof: For even *n*, consider an orientation D_1 of *G* such that v_i is adjacent to v_{i+1}, v_{i-1} where i is odd, $1 \le i \le n$. v_j is adjacent to v_c for all even *j*, $2 \le j \le n$, and v_c is adjacent to v_i for all odd *i*, $1 \le i \le n$. Then D_1 is a strong orientation of *G*. Consider a labeling *C* which ranks $C(v_c) = 3$, $C(v_i) = 1$ for all $v_i \in V_n$, i is odd; and $C(v_j) = 2$ for all $v_j \in V_n$, j is even. Then *C* is a directed node ranking labeling of D_1 with maximum rank 3, which implies $rank(G) \le \chi_r'(D_1) \le 3$. Since *G* contains K_3 as a subgraph, by lemma 4 and proposition 2, we have $rank(G) \ge \chi_r'(K_3) = 3$ which give us $rank(G) = 3$ for even *n*.

For odd n, consider an orientation D_2 of *G* such that v_i is adjacent to v_{i+1} and v_{i-1} where i is odd and $1 \le i \le n$. v_j is adjacent to v_c for all even *j*, $2 \le j < n$, and v_c is adjacent to v_i for all odd *i*, $1 \le i < n$, and the edge $v_1 v_n$ may be oriented either way. Then D_2 is a strong orientation of *G*. Consider a labeling *C* which ranks $C(v_c) = 4$ $C(v_n) = 3$, $C(v_i) = 1$ for all $v_i \in V_n$, $i < n$ and i is odd; and $C(v_j) = 2$ for all $v_j \in V_n$, j is even. Then *C* is a node ranking labeling of D_2 with maximum rank 4, which implies $rank(G) \le \chi_r'(D_2) \le 4$. Since $|V_n|$ is odd, and the adjacent vertices can not have the same rank, there are at least 3 ranks have to be used in V_n. Since V_c can not have the same rank with any vertex in V_n, a directed node ranking labeling of any orientation of *G*, must use at least 4 ranks. Then we have $rank(G) \ge 4$, hence $rank(G) = 4$. ■

3 Conclusion

The node ranking problem of graphs may be extended to digraphs. This paper defined the directed node ranking number and established both upper and lower node ranking number of wheel graphs among all strong orientations.

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